

Capture the Flag (arena2)

William's new web portal `arenafreaks.net` is struggling to take off. As the large bill from the virtual machines provider is expected to arrive sooner and sooner, William is getting frenzied by the addition of new features to the portal, and in particular to its flagship: the game database. As a matter of fact, a couple of days ago William added a database section for *capture-the-flag* games, as usual transcribed in *arenaic notation*.



Figure 1: The earliest known reportage of a *capture-the-flag* game (E. Delacroix, 1830).

The transcription of a capture-the-flag game between N teams in arenaic notation consists in a series of E events, each described by two integers X_i and Y_i from 0 to $N - 1$ representing:

- if $X_i \neq Y_i$, that team X_i stole Y_i 's flag and is running to score a point;
- if $X_i = Y_i$, that team X_i recovered his flag (from the last team which stole it) preventing them from scoring a point.

Help William calculate the final score of each game given its transcription in arenaic notation, assuming that each point that is not later prevented is eventually scored.

🔗 Among the attachments of this task you may find a template file `arena2.*` with a sample incomplete implementation.

Input

The first line contains the two integers N and E . The following E lines contains integers X_i, Y_i .

Output

You need to write a single line with N integer: the score of each team in the match.

Constraints

- $2 \leq N \leq 10\,000$.
- $1 \leq E \leq 1\,000\,000$.
- $0 \leq X_i, Y_i \leq N - 1$ for each $i = 0 \dots N - 1$.
- The events are consistent, that is, there is no “flag recover” event without a “flag conquer” event about the same flag preceding it (and after the previous “flag recover” event).

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** [5 points]: Examples.
- **Subtask 2** [25 points]: $N = 2$.
- **Subtask 3** [40 points]: $N \leq 10$.
- **Subtask 4** [30 points]: No additional limitations.

Examples

input.txt	output.txt
3 4 0 2 1 2 2 0 2 2	1 0 1
2 7 0 1 1 0 1 1 0 0 1 0 0 1 0 1	2 1

Explanation

In the **first sample case**, the second event (team 1 captures flag 2) is prevented by the last event (team 2 recovers flag 2), thus only teams 0 and 2 score a point.

In the **second sample case**, the first event is prevented by the third and the second event by the fourth. The last three events determine the final 2 – 1 score.